Selected Tree Queries and LDBC SNB SF1

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F1K	Q_{desc}	Root	441.595	2.265	2.276	194.965x	194.022x	6.348	3.089	2.515	2.055x	2.524x	29.528	28.457	41.476	1.038x	0.712x
F1K	Q_{desc}	Leaf parent	13.020	1.950	1.828	6.677x	7.123x	3.564	2.043	1.963	1.744x	1.816x	1.776	1.725	1.876	1.030x	0.947x
WT100	Q_{desc}	Root	25.971	2.194	1.808	11.837x	14.364x	3.474	1.535	1.506	2.263x	2.307x	11.628	13.204	13.506	0.881x	0.861x
WT100	Q_{desc}	Leaf parent	11.713	1.749	1.724	6.697x	6.794x	3.020	1.499	1.545	2.015x	1.955x	1.732	2.101	1.831	0.824x	0.946x
WT1K	Q_{desc}	Root	1,152.546	2.632	2.736	437.897x	421.252x	8.740	1.642	2.225	5.323x	3.928x	58.822	45.133	46.768	1.303x	1.258x
WT1K	Q_{desc}	Leaf parent	21.868	2.260	1.738	9.676x	12.582x	3.498	1.597	1.714	2.190x	2.041x	2.180	2.340	2.475	0.932x	0.881x
WT10K	Q_{desc}	Root	112,250.550	8.642	10.905	12,988.955x	10,293.494x	24.882	10.290	8.581	2.418x	2.900x	407.507	486.496	503.709	0.838x	0.809x
WT10K	Q_{desc}	Leaf parent	114.708	5.704	1.900	20.110x	60.373x	2.958	4.574	2.867	0.647x	1.032x	2.413	2.628	2.598	0.918x	0.929x
WT100K	Q_{desc}	Root	23,750,824.047	76.364	74.205	311,021.215x	320,070.400x	122.530	91.594	75.554	1.338x	1.622x	4,312.966	5,124.128	5,269.493	0.842x	0.818x
WT100K	Q_{desc}	Leaf parent	575.098	37.432	1.945	15.364x	295.680x	2.223	50.690	26.894	0.044x	0.083x	1.210	1.306	1.312	0.926x	0.922x
SNB/C	Q_{desc}	Root	91,997.459	680.986	1.951	135.094x	47,154.003x	3.569	416.958	391.391	0.009x	0.009x	4.709	4.490	4.710	1.049x	1.000x
SNB/C	Q_{desc}	Leaf parent	22,390.973	724.077	2.021	30.923x	11,079.155x	3.436	436.106	392.928	0.008x	0.009x	1.578	1.577	1.590	1.001x	0.992x
SNB/P	Q_{desc}	Root	16,838.769	2.706	2.559	6,222.753x	6,580.215x	6.908	2.499	2.252	2.764x	3.067x	46.343	41.164	42.907	1.126x	1.080x
SNB/P	Q_{desc}	Leaf parent	14,506.638	1.606	1.858	9,032.776x	7,807.663x	3.830	2.114	1.876	1.812x	2.042x	1.132	1.304	1.533	0.868x	0.738x
SNB/T	Q_{desc}	Root	14,465.288	1.870	1.900	7,735.448x	7,613.309x	3.337	1.704	1.613	1.958x	2.069x	9.695	10.958	11.340	0.885x	0.855x
SNB/T	Q_{desc}	Leaf parent	14,497.296	1.605	1.684	9,032.583x	8,608.846x	2.943	1.588	1.507	1.853x	1.953x	1.530	1.521	1.535	1.006x	0.997x
F1K	$Q_{a\&d}$	Root–leaf	12.471	1.419	1.477	8.789x	8.443x	19.079	1.005	0.911	18.984x	20.943x	1.730	1.137	1.551	1.522x	1.115x
F1K	$Q_{a\&d}$	Deep parent–leaf	12.246	1.628	1.455	7.522x	8.416x	21.627	0.996	0.920	21.714x	23.508x	1.849	1.172	1.557	1.578x	1.188x
WT100	$Q_{a\&d}$	Root–leaf	11.364	1.706	1.602	6.661x	7.094x	19.110	0.917	0.941	20.840x	20.308x	1.384	0.988	1.103	1.401x	1.255x
WT100	$Q_{a\&d}$	Deep parent–leaf	11.087	1.473	1.465	7.527x	7.568x	18.289	0.899	0.943	20.344x	19.394x	1.225	1.098	1.255	1.116x	0.976x
WT1K	$Q_{a\&d}$	Root–leaf	12.688	1.564	1.534	8.113x	8.271x	20.703	0.844	0.891	24.530x	23.236x	1.060	0.805	1.127	1.317x	0.941x
WT1K	$Q_{a\&d}$	Deep parent–leaf	12.054	1.544	1.509	7.807x	7.988x	18.594	0.850	0.885	21.875x	21.010x	1.318	1.283	1.324	1.027x	0.995x
WT10K	$Q_{a\&d}$	Root–leaf	19.108	1.724	1.774	11.084x	10.771x	21.210	0.860	0.859	24.663x	24.692x	1.386	1.200	0.910	1.155x	1.523x
WT10K	$Q_{a\&d}$	Deep parent–leaf	18.490	1.753	1.732	10.548x	10.676x	21.004	0.959	0.929	21.902x	22.609x	1.230	1.121	1.136	1.097x	1.083x
WT100K	$Q_{a\&d}$	Root–leaf	79.819	1.898	1.811	42.054x	44.075x	20.214	0.937	0.876	21.573x	23.075x	1.577	1.298	1.294	1.215x	1.219x
WT100K	$Q_{a\&d}$	Deep parent–leaf	81.240	1.833	1.790	44.321x	45.385x	22.522	0.844	0.892	26.685x	25.249x	1.282	1.261	1.329	1.017x	0.965x
SNB/C	$Q_{a\&d}$	Root–leaf	14,648.588	2.020	1.860	7,251.776x	7,875.585x	22.268	1.035	0.926	21.515x	24.048x	1.557	1.190	1.084	1.308x	1.436x
SNB/C	$Q_{a\&d}$	Deep parent–leaf	14,582.521	2.119	1.814	6,881.794x	8,038.876x	19.387	1.274	1.007	15.217x	19.252x	1.161	1.063	1.126	1.092x	1.031x
SNB/P	$Q_{a\&d}$	Root–leaf	14,525.595	1.120	1.493	12,969.281x	9,729.133x	19.577	0.801	0.892	24.441x	21.947x	1.421	1.116	1.079	1.273x	1.317x
SNB/P	$Q_{a\&d}$	Deep parent–leaf	14,422.871	1.565	1.481	9,215.892x	9,738.603x	17.314	0.882	0.908	19.630x	19.068x	1.267	1.097	1.103	1.155x	1.149x
SNB/T	$Q_{a\&d}$	Root–leaf	14,538.547	1.608	1.687	9,041.385x	8,617.989x	18.943	0.824	0.946	22.989x	20.024x	1.204	1.146	1.200	1.051x	1.003x
SNB/T	$Q_{a\&d}$	Deep parent–leaf	14,437.473	1.439	1.679	10,032.990x	8,598.852x	20.531	0.832	0.876	24.677x	23.437x	1.389	1.109	1.091	1.252x	1.273x
SNB	IC12	–	188,529.785	170,910.577	173,980.686	1.103x	1.084x	396.524	304.851	403.867	1.301x	0.982x	157.466	119.408	106.318	1.319x	1.481x
SNB	IS2	–	61,383.859	3,842.891	13,060.742	15.973x	4.700x	43.709	242.223	277.390	0.180x	0.158x	4.510	4.535	10.652	0.994x	0.423x
SNB	IS6	–	17,296.910	2,172.181	3,269.589	7.963x	5.290x	28.813	22.864	118.383	1.260x	0.243x	1.897	2.004	1.817	0.947x	1.044x

Queries: Q_{desc} = All Descendants; $Q_{a\&d}$ = Check if Ancestor; IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.
Parameters: Root = root of largest tree; Leaf parent = parent of a leaf; Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. All times are in ms.
Graphs: F = forest; WT = ultrawide tree; K = 1,000 nodes; SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color:

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup